BACKPLANE / CONTINUOUS–WIRE SCHEMATIC

137 original components | Actual pin numbers | No electrical net labels | 20 Oct 2026

Dots join wires. Crossings without dots do not join. Blue captions are plain text only.

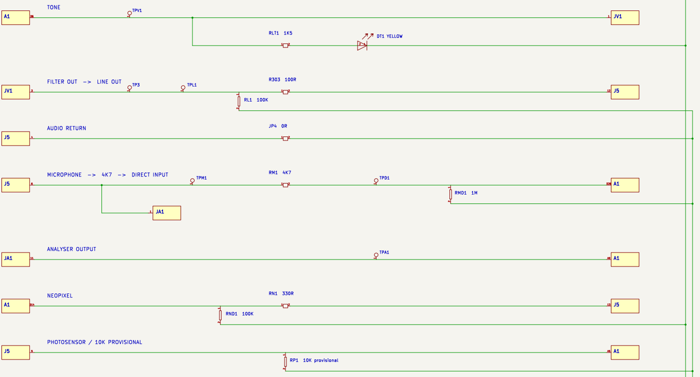
Repeated connector references are contacts of the SAME physical connector. This is a logical wiring drawing.

A1 = Mega–pattern controller socket | JA1 = analyser socket | JV1–JV5 = voice sockets | J5/J1/A–J3/B–J4 = harness

GND
AGND
+5V
MEGA_5V
+12V
dxl_data
analyser/strobe
analyser/reset
amp shutdown

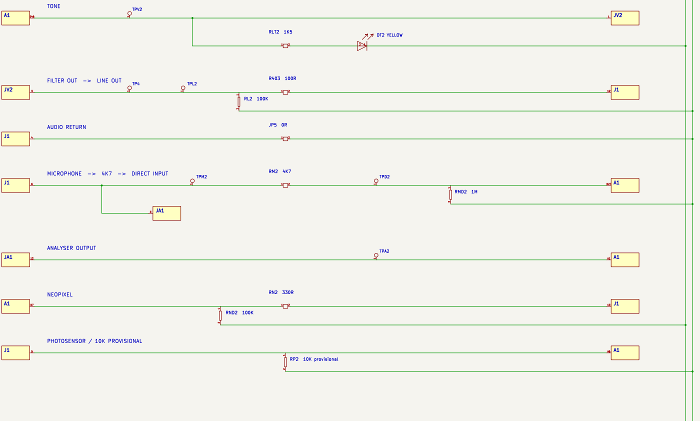
01 FEMALE1 / SIGNAL PATHS

CONTROLLER / SOURCE **EXISTING CONDITIONING / TEST PADS** **CARD / HARNESS CONTACT**

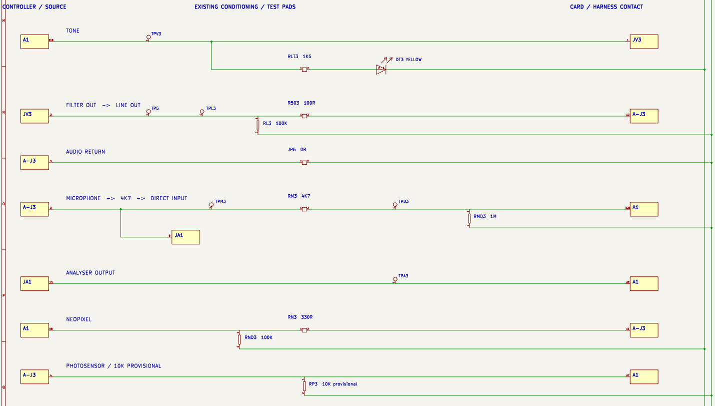

TONE
A1
TPV1
JV1
RLT1 1K5
DT1 YELLOW
FILTER OUT –> LINE OUT
JV1
TP3
TPL1
R303 100R
RL1 100K
J5
AUDIO RETURN
J5
JP4 0R
MICROPHONE –> 4K7 –> DIRECT INPUT
J5
TPM1
RM1 4K7
TPD1
RMD1 1M
A1
JA1
ANALYSER OUTPUT
JA1
TPA1
A1
NEOPIXEL
A1
RN1 330R
RND1 100K
J5
PHOTOSENSOR / 10K PROVISIONAL
J5
RP1 10K provisional
A1

02 FEMALE2 / SIGNAL PATHS

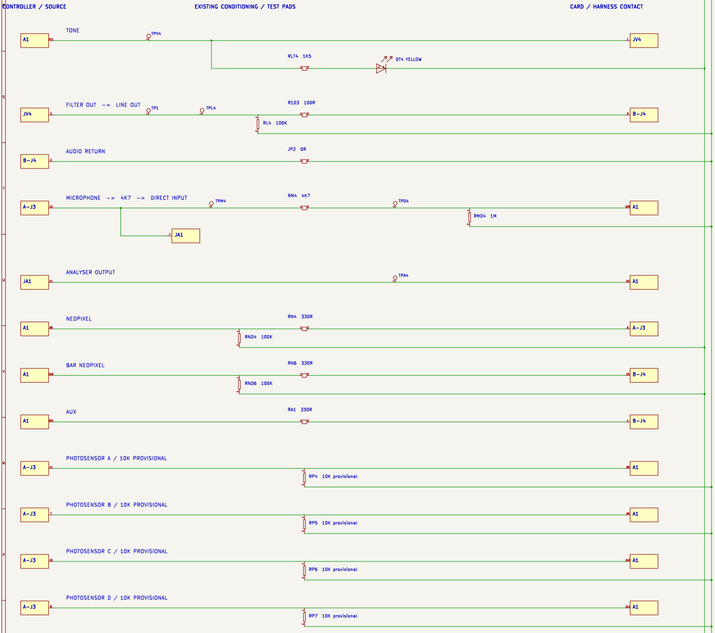
CONTROLLER / SOURCE **EXISTING CONDITIONING / TEST PADS** **CARD / HARNESS CONTACT**


TONE
A1
TPV2
JV2
RLT2 1K5
DT2 YELLOW
FILTER OUT –> LINE OUT
JV2
TP4
TPL2
R403 100R
RL2 100K
J1
AUDIO RETURN
J1
JP5 0R
MICROPHONE –> 4K7 –> DIRECT INPUT
J1
TPM2
RM2 4K7
TPD2
RMD2 1M
A1
JA1
ANALYSER OUTPUT
JA1
TPA2
A1
NEOPIXEL
A1
RN2 330R
RND2 100K
J1
PHOTOSENSOR / 10K PROVISIONAL
J1
RP2 10K provisional
A1

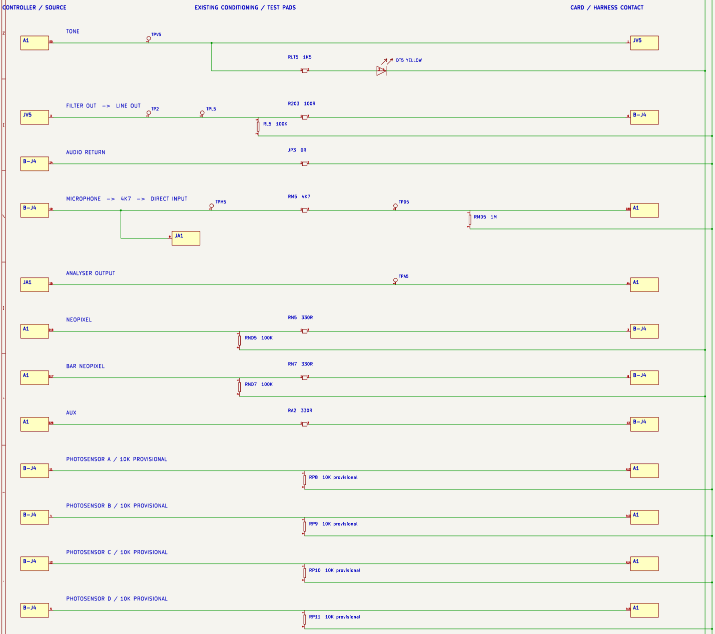
03 FEMALE3 / SIGNAL PATHS


CONTROLLER / SOURCE
EXISTING CONDITIONING / TEST PADS
CARD / HARNESS CONTACT
A1
TONE
TPV3
JV3
RLT3 1K5
DT3 YELLOW
JV3
FILTER OUT -> LINE OUT
TP5
TPL3
R503 100R
RL3 100K
A–J3
A–J3
AUDIO RETURN
JP6 0R
A–J3
MICROPHONE -> 4K7 -> DIRECT INPUT
TPM3
RM3 4K7
TPD3
RMD3 1M
A1
JA1
JA1
ANALYSER OUTPUT
TPA3
A1
A1
NEOPIXEL
RN3 330R
RND3 100K
A–J3
A–J3
PHOTOSENSOR / 10K PROVISIONAL
RP3 10K provisional
A1

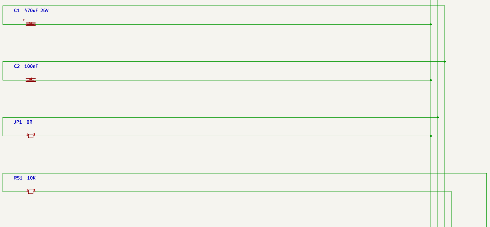
04 MALE1 / SIGNAL PATHS

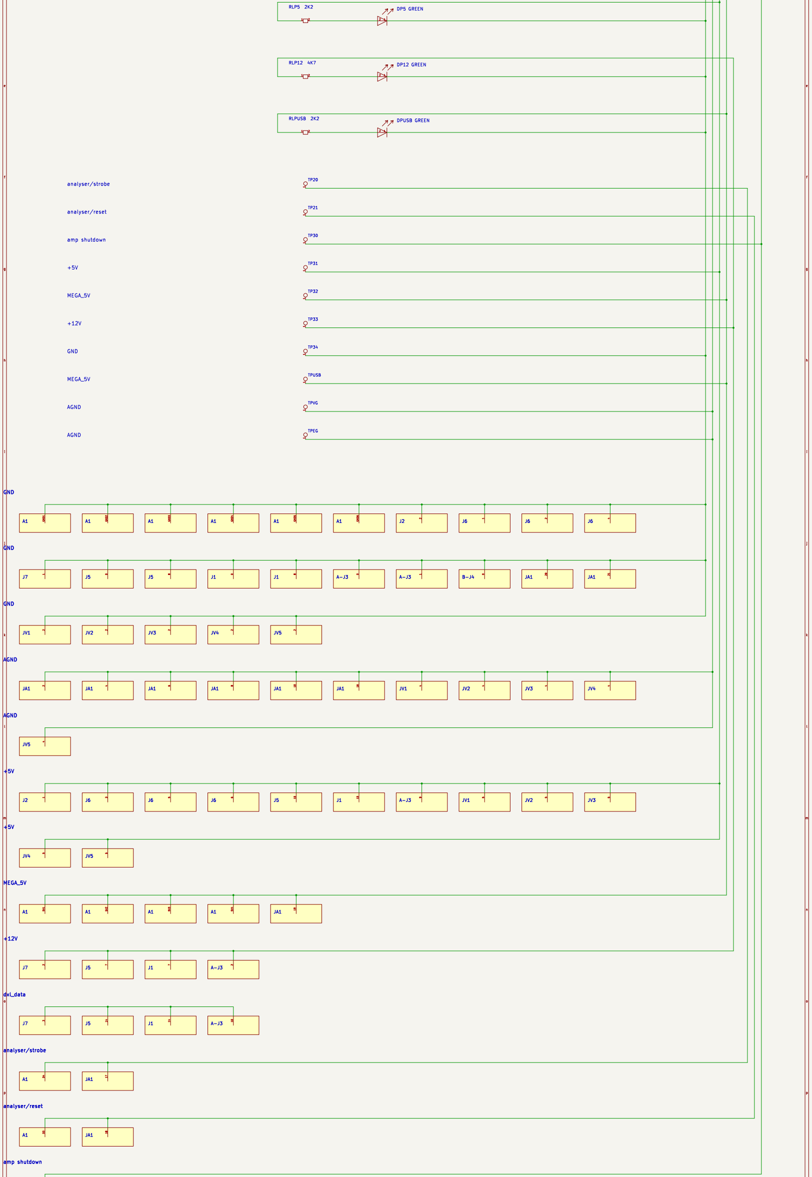

CONTROLLER / SOURCE
EXISTING CONDITIONING / TEST PADS
CARD / HARNESS CONTACT
A1
TONE
TPV4
JV4
RLT4 1K5
DT4 YELLOW
JV4
FILTER OUT -> LINE OUT
TP1
TPL4
R103 100R
RL4 100K
B–J4
B–J4
AUDIO RETURN
JP2 0R
A–J3
MICROPHONE -> 4K7 -> DIRECT INPUT
TPM4
RM4 4K7
TPD4
RMD4 1M
A1
JA1
JA1
ANALYSER OUTPUT
TPA4
A1
A1
NEOPIXEL
RN4 330R
RND4 100K
A–J3
A1
BAR NEOPIXEL
RN6 330R
RND6 100K
B–J4
A1
AUX
RA1 330R
B–J4
A–J3
PHOTOSENSOR A / 10K PROVISIONAL
RP4 10K provisional
A1
A–J3
PHOTOSENSOR B / 10K PROVISIONAL
RP5 10K provisional
A1
A–J3
PHOTOSENSOR C / 10K PROVISIONAL
RP6 10K provisional
A1
A–J3
PHOTOSENSOR D / 10K PROVISIONAL
RP7 10K provisional
A1

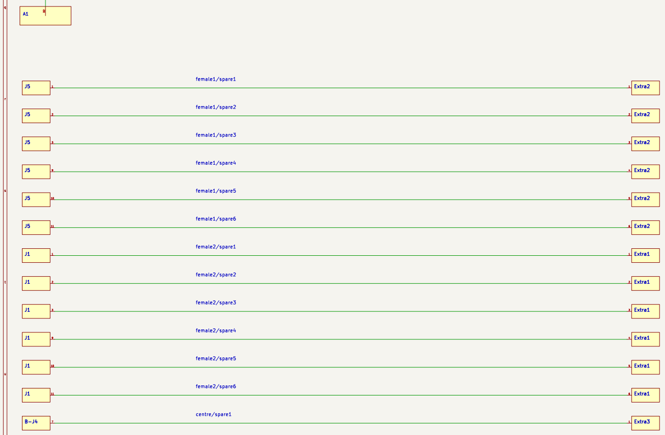
05 MALE2 / SIGNAL PATHS


CONTROLLER / SOURCE
EXISTING CONDITIONING / TEST PADS
CARD / HARNESS CONTACT
TONE
A1
TPV5
JV5
RLT5 1K5
DT5 YELLOW
FILTER OUT -> LINE OUT
JV5
TP2
TPL5
R203 100R
RL5 100K
B-J4
AUDIO RETURN
B-J4
JP3 0R
MICROPHONE -> 4K7 -> DIRECT INPUT
B-J4
TPM5
RM5 4K7
TPD5
RMD5 1M
A1
JA1
ANALYSER OUTPUT
JA1
TPA5
A1
NEOPIXEL
A1
RN5 330R
RND5 100K
B-J4
BAR NEOPIXEL
A1
RN7 330R
RND7 100K
B-J4
AUX
A1
RA2 330R
B-J4
PHOTOSENSOR A / 10K PROVISIONAL
B-J4
RP8 10K provisional
A1
PHOTOSENSOR B / 10K PROVISIONAL
B-J4
RP9 10K provisional
A1
PHOTOSENSOR C / 10K PROVISIONAL
B-J4
RP10 10K provisional
A1
PHOTOSENSOR D / 10K PROVISIONAL
B-J4
RP11 10K provisional
A1

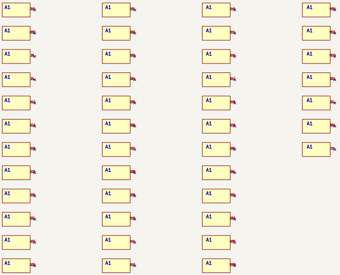
06 POWER / SHARED CONTROL / SPARES


C1 470uF 25V
C2 100nF
JP1 0R
RS1 10K


RLP5 2K2
DP5 GREEN
RLP12 4K7
DP12 GREEN
RLPUSB 2K2
DPUSB GREEN
analyser/strobe
TP20
analyser/reset
TP21
amp shutdown
TP30
+5V
TP31
MEGA_5V
TP32
+12V
TP33
GND
TP34
MEGA_5V
TPUSB
AGND
TPVG
AGND
TPEG
GND
A1
A1
A1
A1
A1
A1
J2
J6
J6
J6
GND
J7
J5
J5
J1
J1
A–J3
A–J3
B–J4
JA1
JA1
GND
JV1
JV2
JV3
JV4
JV5
AGND
JA1
JA1
JA1
JA1
JA1
JA1
JV1
JV2
JV3
JV4
AGND
JV5
+5V
J2
J6
J6
J6
J5
J1
A–J3
JV1
JV2
JV3
+5V
JV4
JV5
MEGA_5V
A1
A1
A1
A1
JA1
+12V
J7
J5
J1
A–J3
dxl_data
J7
J5
J1
A–J3
analyser/strobe
A1
JA1
analyser/reset
A1
JA1
amp shutdown


A1
J5 1 female1/spare1 1 Extra2
J5 2 female1/spare2 2 Extra2
J5 3 female1/spare3 3 Extra2
J5 9 female1/spare4 4 Extra2
J5 10 female1/spare5 5 Extra2
J5 11 female1/spare6 6 Extra2
J1 1 female2/spare1 1 Extra1
J1 2 female2/spare2 2 Extra1
J1 3 female2/spare3 3 Extra1
J1 9 female2/spare4 4 Extra1
J1 10 female2/spare5 5 Extra1
J1 11 female2/spare6 6 Extra1
B-J4 7 centre/spare1 1 Extra3

07 UNUSED CONTACTS / MECHANICAL ITEMS


A1
A1
A1
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A1
A1
A1

M1 / U2D2

HV1 / M3 card retention

HV2 / M3 card retention

HV3 / M3 card retention

HV4 / M3 card retention

HV5 / M3 card retention

HA1 / M3 analyser retention

HA2 / M3 analyser retention

amp shutdown
analyser/reset
analyser/strobe
dxl_data
+12V
MEGA_5V
+5V
AGND
GND

Reading copy: original components and connections preserved
Colloquy of Mobiles
Sheet: /
File: backplane-wired.kicad_sch
Title: Backplane / continuous wiring / reading copy
Size: User
Date: 2026-10-20
Rev: A wired copy
KiCad E.D.A. 9.0.7
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